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| <b>Q5</b>       |                                                                                                                                                                                                                                                                          |                                             |
| <b>(a)(i)</b>   | $Q = \text{mass} \times \text{specific latent heat of vaporisation}$<br>$= 0.37 \times 2.3 \times 10^6$<br>$= 8.5 \times 10^5 \text{ J}$                                                                                                                                 | <b>A1</b>                                   |
| <b>(a)(ii)</b>  | $pV = nRT$<br>$T = (100 + 273) = 373 \text{ K}$<br>Number of moles, $n = 0.37 \times 1000 \text{ g} / 18 \text{ g}$<br>$V = \frac{(0.37 \times 1000) \times 8.31 \times 373}{18 \times (1.0 \times 10^5)}$ $= 0.63714$<br>$= 0.64 \text{ m}^3$                           | <b>B1</b><br><b>B1</b><br><b>B1</b>         |
| <b>(a)(iii)</b> | Work done <i>by</i> the water $= (\text{atmospheric pressure})(\text{increase in volume})$<br>$= (1.0 \times 10^5)(0.64)$<br>$= 6.4 \times 10^4 \text{ J}$                                                                                                               | <b>A1</b>                                   |
| <b>(a)(iv)</b>  | Work done on water is negative.<br>From the first law of thermodynamics,<br>increase in internal energy = heat supplied + work done <i>on</i> water<br>$= (8.5 - 0.64) \times 10^5$<br>$= 7.9 \times 10^5 \text{ J}$                                                     | <b>M1</b><br><b>A1</b>                      |
| <b>(b)</b>      | Kinetic energy of the molecules remains unchanged because there is no temperature change.<br>Potential energy of the molecules increases, because molecular bonds are broken and the molecules are further apart.<br>Hence, the internal energy of the system increases. | <b>B1</b><br><br><b>B1</b><br><br><b>A0</b> |

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